

Zexuan Kang

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EDUCATION BACKGROUND

The Chinese University of Hong Kong, Shenzhen(CUHKSZ), China 09/2022 - Expected. 06/2026
Bachelor of Science in Data Science and Big Data Technology | GPA: 3.4/4.0

Core Coursework: Introduction to Data Science, Data Structures, Optimization, Machine Learning, Programming Methodology(Python/C/C++), Design and Analysis of Algorithms, Cloud Computing, Introduction to Quantum Computation and Quantum Information, Introduction to Coding Theory, Calculus I&II, Linear Algebra, Probability, Discrete Mathematics, Stochastic Processes, Stochastic Simulation, Investment Analysis and Portfolio Management.

WORK EXPERIENCES

Li Auto Inc., *Algorithm Engineer Intern, China* 06/2025 -08/2025

- Developed productivity tools leveraging large language models (LLMs) to automate code generation for in-house automotive software development, significantly enhancing R&D efficiency in a vertical domain.
- Conducted domain-specific code demand analysis and independently integrated LLMs with automotive software APIs, achieving 95% accuracy in code outputs and deploying solutions on internal GPU clusters.
- Coordinated across research and development teams for data annotation and agile development, ensuring scalable and reliable deployment of LLM-based tools; authored internal technical documentation and led seminars and training sessions to promote adoption and responsible use of LLM applications across engineering departments.

Shanghai Baoshu Tech Co., Ltd., *Product R&D Intern, China* 05/2023 -08/2023

- Contributed to a core robotics R&D project, programmed the intelligent teaching robot movements, and designed the demonstration workflow, ensuring smooth deployment in real-world applications.
- Built and applied a proprietary corpus for "Hualong N.O. 1" (a nuclear power unit); responded to data annotation and algorithm testing; incorporated deep learning using TensorFlow and PyTorch to fine-tune model performance, achieving notable improvements in accuracy and efficiency.

RESEARCH EXPERIENCE

Anchored Densest Subgraph Problem (ADSS) on Directed Graphs 09/2024 – 01/2025

- Identified a gap in conventional densest subgraph research where key anchor structures are often overlooked. Initiated a study on the Anchored Densest Subgraph (ADSS) problem, which aims to find a subgraph with the highest average edge density while containing a predefined target subgraph.
- Derived a dynamic programming formulation from the linear programming model and applied the Frank-Wolfe algorithm to optimize the solution under structural constraints. Implemented the algorithm and developed a quality evaluation framework to validate its effectiveness on real-directed graph data.
- Preliminary results demonstrate that the proposed approach effectively balances density maximization and constraint satisfaction with significant improvements in computational efficiency.

Approximate Maximum Clique Problem in Bipartite Graphs 09/2023- 05/2024

- Conducted research on the defective-clique problem in bipartite graphs, inspired by gaps found in existing literature on network clustering.
- Independently developed and optimized algorithms using alpha-beta core compression and ordering techniques. These optimizations reduced computation time by 10%, significantly accelerating the clique-finding process and offering practical insight into solving defective clique problems in bipartite settings.

Beijing Mobile Customer_Experience Analytics, *MathorCup* 01/2023

- Demonstrated the theoretical equivalence between Pearson correlation and Euclidean distance on standardized data, then quantified links between 50+ network quality variables and overall satisfaction, isolating the top five drivers for both voice and mobile data services.
- Built and tuned XGBoost with feature crossing and regularization; achieved 85% prediction accuracy on potential customer rating sets and auto-generated result sheets required by the competition platform.

SKILLS & CERTIFICATES & HONORS

Programming Languages & Tools: Python, C++, TensorFlow, PyTorch

Algorithmic Proficiency: Graph Theory, Network Flow, Dynamic Programming, Recommender Systems

Honors & Awards: Bronze Medal, ACM-ICPC Regional Contest | First Prize, CUHK(SZ) Algorithm Programming Competition | Second Prize, National Olympiad in Informatics in Provinces (NOIP)